

DATE: 17 Dec. 2022, FN (9 am -12 noon)

Total (4+2 questions)

Common data: Use the following common data if not mentioned specifically in the question

For 0.25 μ m Technology node --

Take $V_{DD} = 2.5$ V, $V_{TON} = 0.4$ V, $V_{TOP} = -0.4$ V, $L_{min} = 250$ nm, $\lambda_n = 0.1$ V $^{-1}$, $\lambda_p = 0.01$ V $^{-1}$, $\mu_n C_{ox} = 120$ μ A/V 2 , $\mu_p C_{ox} = 40$ μ A/V 2 , $g(inv) = 1$, $p(inv) = 1$

- Use long channel approximation and lambda-based design rules unless specially mentioned.
- Use square law current equation unless specifically mentioned. Neglect channel length modulation in the current calculation, if not mentioned specifically. The body of all PMOS is tied to V_{DD} while all NMOS is tied to the ground unless specially mentioned.

Answer the sub-parts of a question in one place and in the order in which they appear.

SECTION - A

Q1. Consider the buffer circuit given in Fig.1. The input V_{in} swings from 0V to 5V and the load capacitance is 10pF. Use long channel MOSFET equations.

The following parameters are given: $\mu_n C_{ox} = 25$ μ A/V 2 ,

$\mu_p C_{ox} = 10$ μ A/V 2 , $V_{ton} = 1.1$ V, $V_{top} = -1$ V and $L = 1$ μ m.

- It is desired to design the 1st stage inverter such that the average power dissipation (Assuming 50% duty cycle) is 2mW and the output low voltage (V_{OL}) is 1V. Find the values of W_p and W_n .
- Using the sizing of the 1st stage inverter, the 2nd stage inverter is to be designed for equal rise and fall times. Use the average current method and find the ratio of $[W_p / W_n]$.

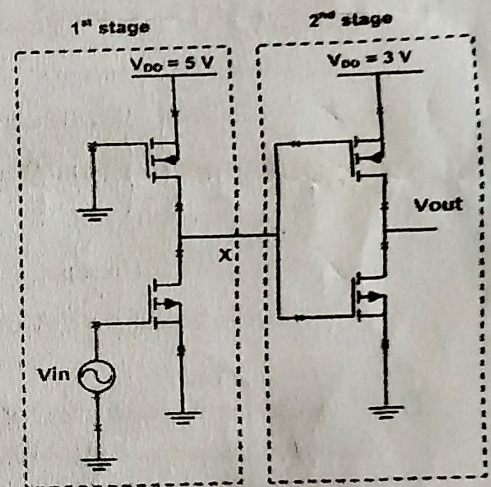


Fig. 1

[18 marks]

Q2. A transmission gate is designed in a technology node such that $R_{eqp} = 90$ k Ω and $R_{eqn} = 10$ k Ω . The capacitance at the output of each transmission gate is 1pF.

- Implement the function $Y = [(ABC + D) E + FH]$ using the transmission gate/s.
- The function block of part (a) is now used in a circuit as shown in Fig. 2. Inputs to each block are different. It is intended to insert buffers to achieve the lowest possible delay. Given the propagation delay of the buffer for the technology is 30ns, mark the nodes where buffers are to be inserted and justify.

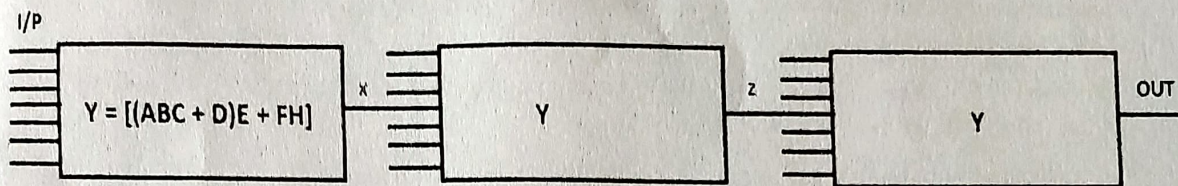


FIG. 2

[7 Marks]

Q3. A resistive load inverter is given with the following parameters:

$$V_{DD} = 2.5 \text{ V}, V_{ton} = 0.4 \text{ V}, \mu_n C_{ox} = 115 \mu\text{A/V}^2.$$

It is given that the highest V_{in} for which the transistor operates in the saturation region is 1.8 V with a 4 mW power supply. Use the long channel MOSFET equation and ignore channel length modulation, body bias effect, and parasitic capacitances.

- Determine R_L . Also, Determine $[W/L]$ for the transistor.
- Determine V_{OL} , V_{OH} , V_{IL} , V_{IH} , V_m , and noise margins. Sketch and label Voltage Transfer Characteristics (VTC).
- Now if $[W/L]$ is increased by 20% while keeping all other parameters unchanged. Calculate new V_m . Sketch a labeled new VTC on the same plot as in part (c).

[15 marks]

Q4. (a) Consider the layout of an NMOS transistor shown in Fig. 3.

The process parameters are $t_{ox} = 0.05 \mu\text{m}$,

Given--- $V_{DD} = 5\text{V}$, $K_{eq} = 0.44$, $K_{eq_{sw}} = 0.46$, $C_{j0} = 0.95$ pico Farad / μm^2 , $C_{j0_{sw}} = 3.5$ pico Farad / μm^2

- Find the large signal equivalent **drain parasitic capacitance**.
- Further, this device is being scaled down to an unknown technology node using constant voltage scaling with a scaling factor of $S=10$. Now, it is observed that this MOS structure has a huge gate leakage current because of the thin gate oxide layer. So, SiO_2 layer is replaced by the Zirconium dioxide (ZrO_2) layer. Now, find the required thickness of ZrO_2 to keep

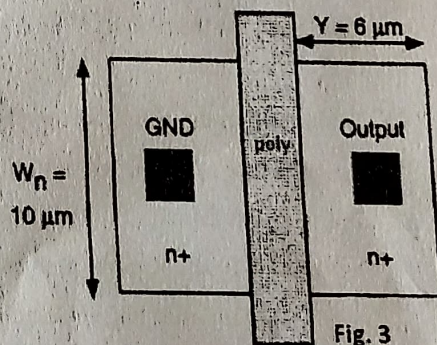


Fig. 3

the **electrical parameters of the scaled NMOS device the same**. Also, find the **maximum electric field** in the new gate oxide. (Consider Dielectric constant of $\text{SiO}_2 = 3.9$, Dielectric constant of $\text{ZrO}_2 = 25$, $V_{gs} = 5\text{V}$, $V_{ds} = 0\text{V}$).

- A gate oxide (SiO_2) layer (for the NMOS in Fig. 3) is grown using a dry oxidation process at 1250°C . Calculate the number of atoms of Silicon consumed in the process.

(Density of $\text{Si} = 2.33 \text{ g/cm}^3$, Atomic mass of silicon = 28, Avogadro's number = 6.023×10^{23} , Molecular weight of silicon dioxide = 60, Surface area of gate oxide = $40 \mu\text{m}^2$).

[15 marks]

Write the section name clearly on top of each sheet.

Use common parameters given in SECTION A

Answer the sub-parts of a question in one place and in the order in which they appear.

SECTION - B

Neglect body effect

Q1. Consider the circuit given in Fig. 1B.

$$V_{od} = V_{out1} - V_{out2}, V_{id} = V_{in}$$

Given $I_{ss} = 200\mu A$, $V_{ov} = 0.2V$, $I_s = 100\mu A$

Matched transistors--

- (M1, M2, M3, M4),
- (M5, M6, M7, M8),
- (M9, M10),
- (M11, M12)

Load capacitance ---

-at node V_1, V_2 is $C_x = 0.1 C_L$

-at node V_{out1}, V_{out2} is $C_L = 100$ pico farad

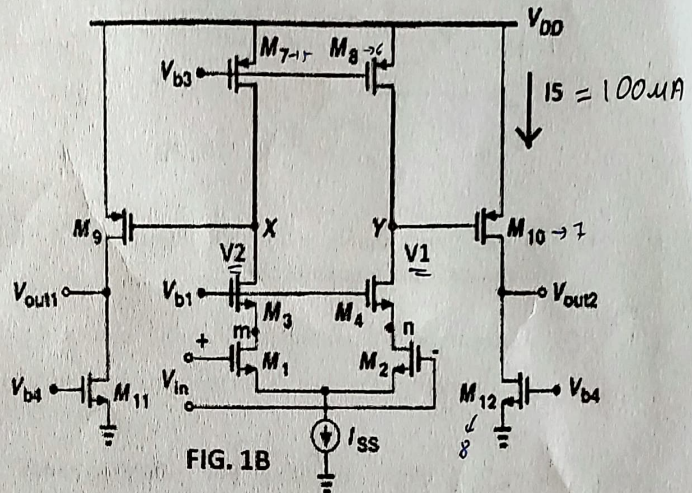


FIG. 1B

- a) Sketch and label low-frequency small signal model, with differential input V_{id} , of Fig. 1B.
- b) Intuitively, determine the maximum small signal gain $\left(\frac{v_{od}}{v_{id}}\right)$. Hence determine the maximum input (peak to peak) which will produce undistorted output.
- c) Determine the full power bandwidth for full swing output.
- d) Determine the value of the additional capacitor required to be added at the output to shift UGB (unity gain bandwidth) to the second dominant pole location.

Consider Fig. 2B--- the load capacitance is as shown in Fig. 3B---

Assume matching wherever necessary

- (M5, M6) have large r_o
- Take $C_L = C_1 = 10$ pico farad, $R_1 = 100 K\Omega$
- $V_{od} = V_{out1} - V_{out2}$, $V_{id} = V_{in}$

Given $g_{m1,2} = 1mA/V$, $V_{ov} = 0.2V$, $r_{on} = r_{op} = 100 M\Omega$

When switch $S1$ is closed,

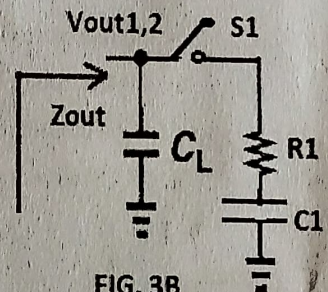


FIG. 3B

- e) Derive the expression for $Z_{out}(s)$
- f) Sketch and label the Bode Magnitude (corner) plot, clearly showing voltage gain with respect to frequency. Neglect pole at node X, Y. (Plot should be not to scale but qualitatively correct)

[25 marks]

